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Software is the collection of programs, procedures and data. Example: ms word, excel etc.

Software engineering is different from system engineering as follows:

SN	Component	Software engineering	System Engineering
1.	Focus	It focuses to design and development of a software system.	It focuses to design a complex and influential systems.
2.	Scope	It has narrow scope	It has broad scope.
3.	Objective	It's objective is to design, develop, test and maintain a software.	It's objective is to design, test and integration of architecture.
4.	Knowledge	It requires skills of programming, algorithms, software pattern.	It requires thinking of system, problem solving capacity, architectural design knowledge.
5.	Complexity	Software design and development is less complex than system design.	System designing is more complex work.

Examples of software engineering:

Development of mobile App, healthcare application, educational app etc.

Examples of System Engineering:

Temperature system, Expert system, E-Governance system etc.

1b

- Software Myths are the false beliefs and superstition related to software development.

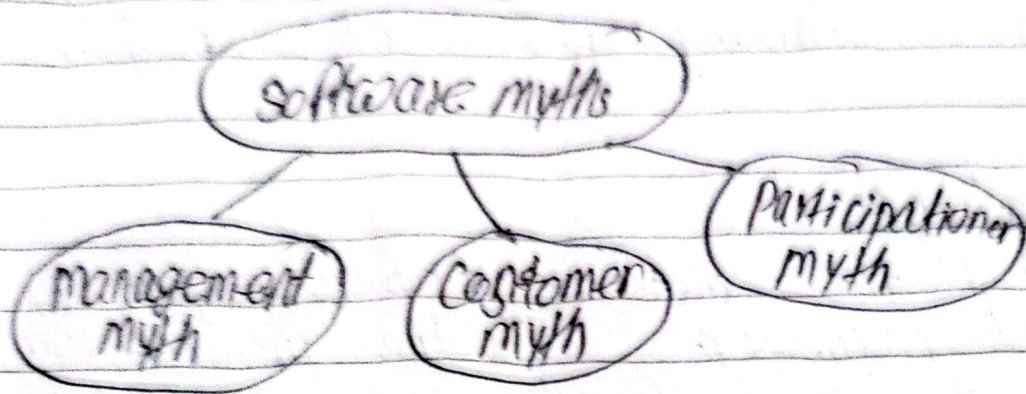
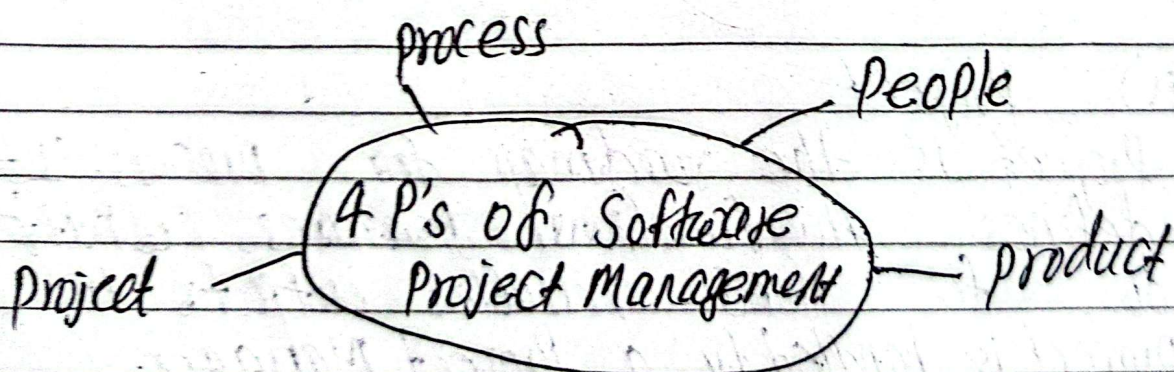


fig: Types of Software myth

- Software myth creates problems in good and high quality software.

→ The 4 P's of Software Project Management are the backbone of managing software development. They are as follows:



i) People
People are the group of persons who involved in software development. A project manager need to proper guide to the cooking people of a project.

ii) Product
Product is the outcome of a project. A project manager fix the product. He/she analyses and decides what will be result for clients. Product guides the project what to do.

iii) Process
Process is the methodology how to develop a software. It defines the steps and way to meet desired outcomes. A project manager shows the path to his/her workers.

iv) Project
Project is the roadmap for process. It defines the functions, features, what to do to find reliable outputs. A project is handled by a Project Manager.

Qa

Software metrics are the quantitative measures to assess software development different aspects

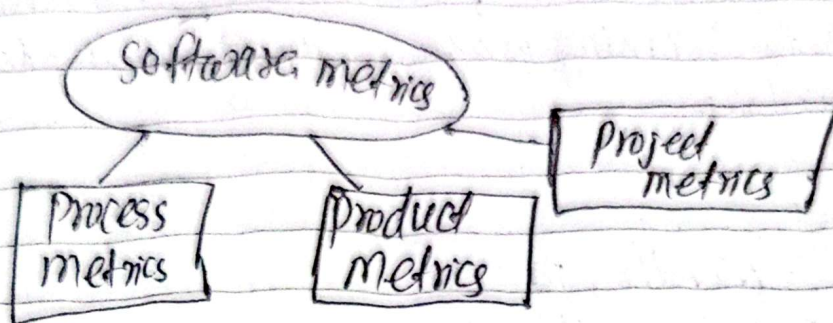


fig: Types of Software metrics

Types of Software metrics:

① Process metrics

- It is the measurement of software development process to find the body of a software.
- It ensures how the process are efficient and reliable.

It can be three types:

- 1) Static process metric: Defined the current process.
Eg: no. of roles.

b) Dynamic process metrics: How process going or define attributes.

c) Origin Evaluation metrics: Define origin to completion processes.

Load time, time cycle, customer satisfaction, throughput, Working Progress measures are the common process metrics.

Advantages:

- define well process
- increases efficiency

① Product Metrics

It is measurement product quality by a project manager during software development.

- It ensures the product is according to customer requirements or not.

Advantages:

- product is clear
- ensure the product meets quality assurance.

(ii) Project Metrics

It measures how project is going.
Checks the project is reliable or not.
It find the details of a project.

Advantages:



- guides the new project
- gives the information about projects
Count the resources, cost, technology
in a project.

2b

Software Measurement Indicators are the technique or tools which measures the different aspects of a software development processes.

Types of Software Measurement Indicators:

- (i) Process Metrics
Measures the process which define the body of software. Developers will be clear how to working during software development.
- (ii) Product Metrics
It indicates the quantity and reliability of a product. It gives description of a product or outcomes.
- (iii) Project Metrics
It measures the project system. How the project completed, how many workers are there. Gives a details about a project.

④ User satisfaction Measurement

It collects feedback from users and show how users are satisfied. A software meets or not consumer's requirements.

⑤ Security Measurement

It checks either the software or not.

It keeps privacy and security or not.

It ensures the reliable and ethical software system.

→ Importance of Software Measurement Indicators:

- It ensure high quality software.
- It quantifies process, product, project and user satisfaction.
- helps to develop reliable and efficient software.
- It tells the requirement to develop a high quality and efficient software system.
- It guides to a project lead and software developer.
- User are informed about the products.

3a

Empirical Project Estimation is the process of estimating the cost of developing software based on previous project and some assumption.

There are two types of Empirical Project Estimation Techniques:

a) Expert Estimation

- It has a group of experts who have a well knowledge about a project.
- In this estimation is done by experts.

- A project manager gather the suggestion of experts and apply it to his/her project estimation.

- Experts analysis the previous projects and predicting future estimation and gives a final estimation.

Advantages

- Expert knowledge is applied.
- nearly meets the estimation.

Disadvantages

- always cannot be true
- finding experts is challenging

b) De-helpi Estimation

- In this a action can be indeed to find estimation where actual estimation is crucial.
- It is a actionable estimation.

• Process of De-helpi estimation

- ① Selection of Experts
- ② initial estimation
- ③ analysis estimation
- ④ review estimation
- ⑤ final estimation

Advantages

- gives actual estimations
- help to manage resources, budget, technology precisely.

Disadvantages

- long time process
- complexity and expensive.

3b

- SDLC stands for Software Development Life Cycle. It is the overall process of defining, designing, coding, testing and deployment of a software system.
- It is a systematic approach of software development process.

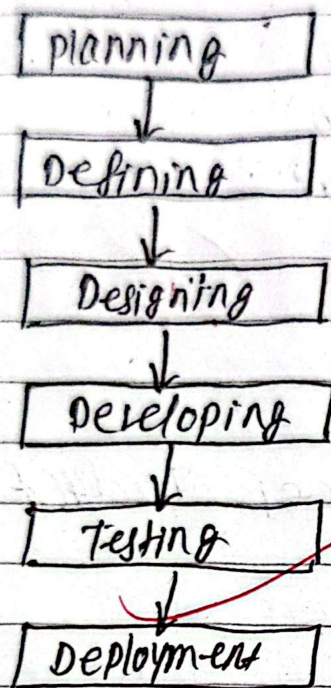


fig: SDLC phases

Phases of SDLC:

1. Planning

It is plans what to develop. In this a developer gathers the requirements from customers using different methods like interviews, survey, brainstorming etc.

2. Defining

It is process of defining resources to develop a software. In this phase feasibility study is done. It is analysed that the software is feasible or not according to economy, technology, budget, legally, resources etc.

3. Designing

It is process of designing the architecture of a software. It is a roadmap or rough for software development.

4. Developing

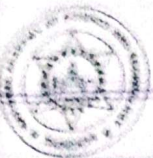
It is the main part a SDLC where a developer develops software. He/she codes the frontend, backend and database.

5. Testing

After coding a tester tests the quality of software. There can be many bugs. So tester test black box test, white box test, unit tests, integration test etc to fix bug and remove them and provides a bugless product.

6. Deployment

After finishing of testing the software is done public. Customer use it. A Developer collects feedback from user and modify the system as user requirements and maintains the software in market.



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4b

~~Ext~~ Xtreme programming is the programming method that provides a high quality product by frequently change, feedback and interactions of software system.

- ~~Life cycle of Xtreme programming~~ defines the process of it.

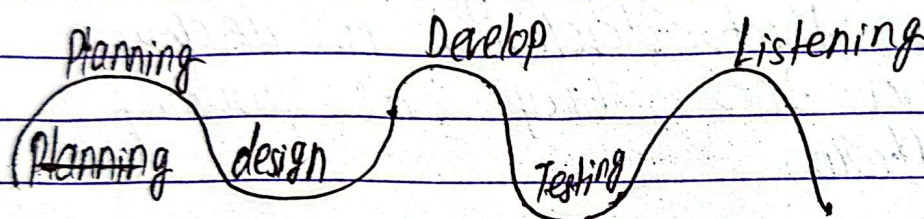


fig: Lifecycle of ~~Ext~~ Xtreme Programming

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Life cycle steps:

a) Planning

In this phase a developer analysis the user requirements and make a plan what to do.

b) Design

After define what to do developer designs the architecture of a software system. This guides to development easily.

c) Development

Based on previous architecture developer codes and develop a software.

d) Testing

A tester checks the bug in the programming that developed a developer. Then remove the bugs as providing reliable software.

e) Listening

After testing the software is public in market and deployed. The customer's feedback are collected and must be modified

The software based on user's demand.

5a

- Scrum is a framework of 'Agile'.
- Scrum is important in Agile methods because:
 - Scrum is a methodology to develop a Agile methods easily and efficiently.
- It is a management framework where user use to self organize tasks and to meet the teams goals.
- It provides platform for all developers at once which makes quick change which is the main aim of agile.
- ~~Scrum~~ is a reliable, efficient and ~~most~~ widely used recent technology.

→ life cycle of Scrum defines the process of Scrum.

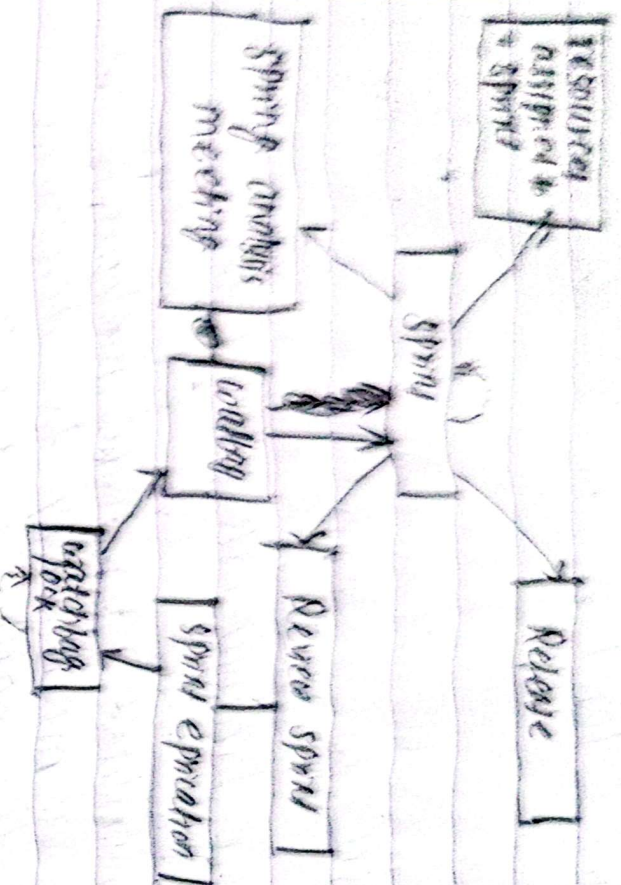


fig: Lifecycle of Scrum

a) Sprint

Sprint is a timebox for scrum. It takes 1 month or more. If previous scrum is completed then it is turn of new one.

b) Release

If the sprint is ~~right~~ and efficient the it will release.

c) Review sprint

If there are some bugs in sprint then it goes to review and checks defects.

d) Sprint expiration

There is analysis of sprint. What todo and What not to do

is defined.

e) Watchbag, lock
It is locked the sprint which need
to watchbag and modify.

f) Watchbag

It is the process to modify sprint
it is defined into two parts: Resource
assigned to ~~sprint~~ and sprint analysis
~~meeting~~.

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5b

Requirement Engineering is the systematic approach to define, analyse and applying the needs of customer of a software system.

It prioritizes necessity of customers and helps to manage it.

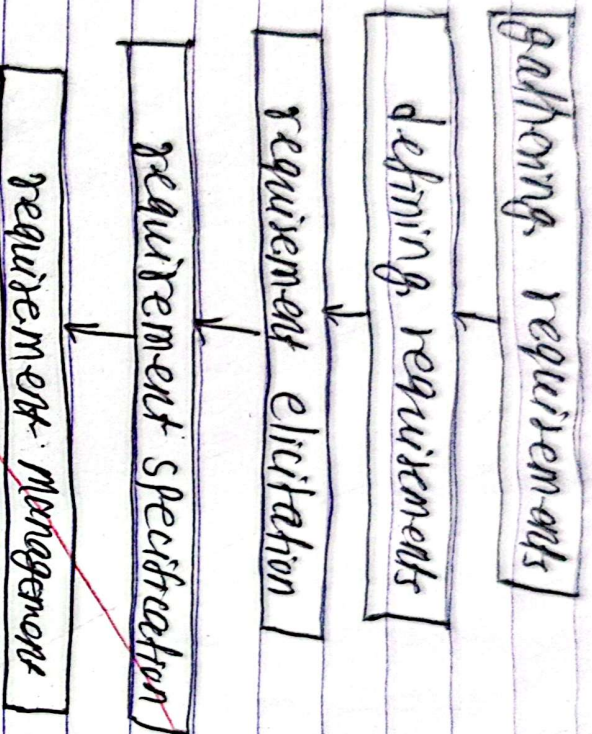


fig: ~~Phases of Requirement Engineering~~

① Gathering Requirement

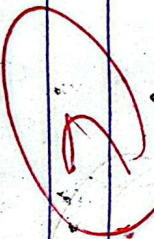
- It is the first phase of requirement engineering in which the needs of customers are gathered in various way like feedback, surveys, interviews etc.

⑩ Defining Requirement
In this phase, an engineer defines the requirement which helps going to fulfill for the customer.

⑪ Requirement elicitation
It is analysis of requirements. How the requirement fulfill, how many resources, budget it takes, time takes, require etc.

⑫ Requirement Specification
It specifies the requirement after analysis. It chooses the best option for fulfill the user's requirements.

⑬ Requirement Management
- In this phase, the requirement of the user are fulfilled. The specialized requirement is attempted to deploy. Which gives satisfaction to user.



fa

OODA stands for Object Oriented Analysis
is the systematic approach to define
objective and modelling of a software
system.

OODA stands for Object Oriented Design
Analysis Design is the process in which
everything is defined as objects as a
real-world entity.

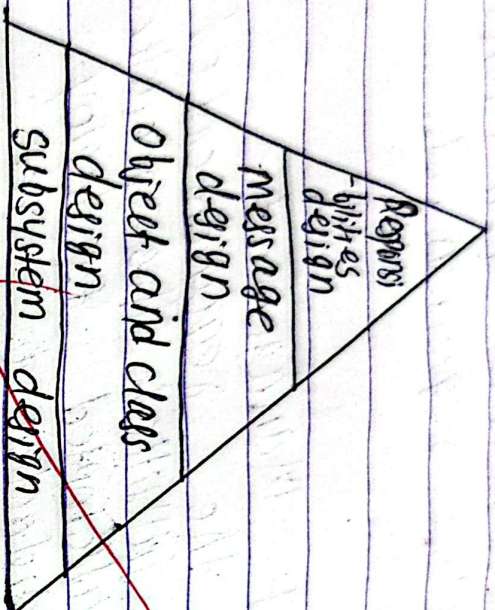


Fig: Process of OODA.

- Process of OODA define how OODA works.

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① Responsibility Design

It defines what is the responsibility of an object or class in a system. It defines what to be done by an object.

② Message design

It is the interconnection of objects. It defines how an object communicates & interacts with other objects.

③ Object and class Design

~~It~~ defines the objects and class in a software. It defines the structure, attributes, behaviours of object and class.

④ Subsystem Design

It is the process of integration of classes and objects in a large unit which acts as a Subsystem Design.

- Advantages of CODD :
- ensuring high quality product
 - support reliability
 - secure and reliable

- Disadvantage of CODD
- more time consuming
 - complex
 - expensive prices

Qa

SDLC methods are:

- waterfall method
- Agile method
- Incremental "
- Iterative "
- V-model
- Big-Bang model
- RAD model
- Prototyping model
- spiral model

a) Waterfall Model

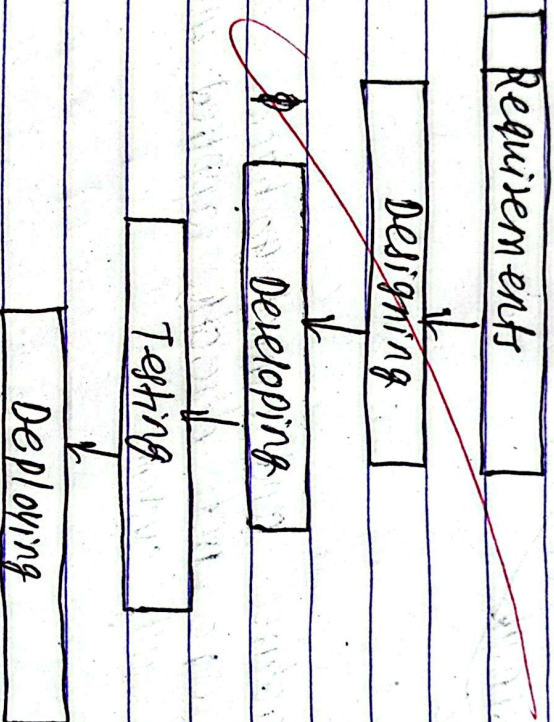


fig: Waterfall Model

- Waterfall model is very simple model.
- It is default traditional model for software development.

Uses:

- small priced
- where requirement are fixed

Steps:

Requirement: gathers the need of customer
Designing: make architecture of software
Developing: coding the software
Testing: identify bugs and remove them
Deploying: public to market and modify as required.

Advantages

- fast and simple
- cost effective

Disadvantages

- linear structure and cannot go back
- changes are not directly applied needed to modify all systems

b) Agile method
- It is today's most famous method
which accept frequent test.

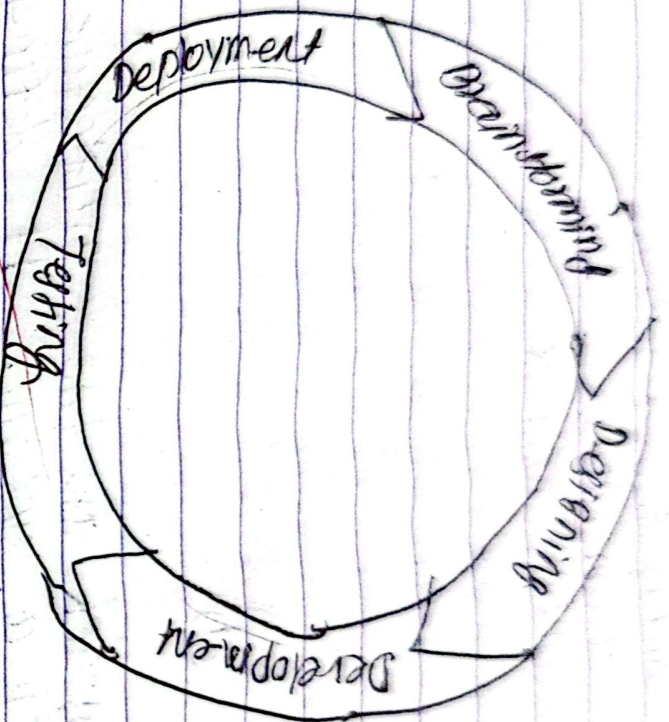


fig: Agile Method

Uses:

- where requirements are rapidly changed

Steps:

Brainstorming: gathering needs by thinking and user feedback.

Designing: making architecture

Developing: writing codes

Testing: eliminating bugs

Deployment: public to user and keep maintain

→ Advantages are frequently acceptable
→ quick access

Disadvantages

- no documentation
- costly and complex

Qb

System model are the architecture or model of a system. It defines the design of a system.

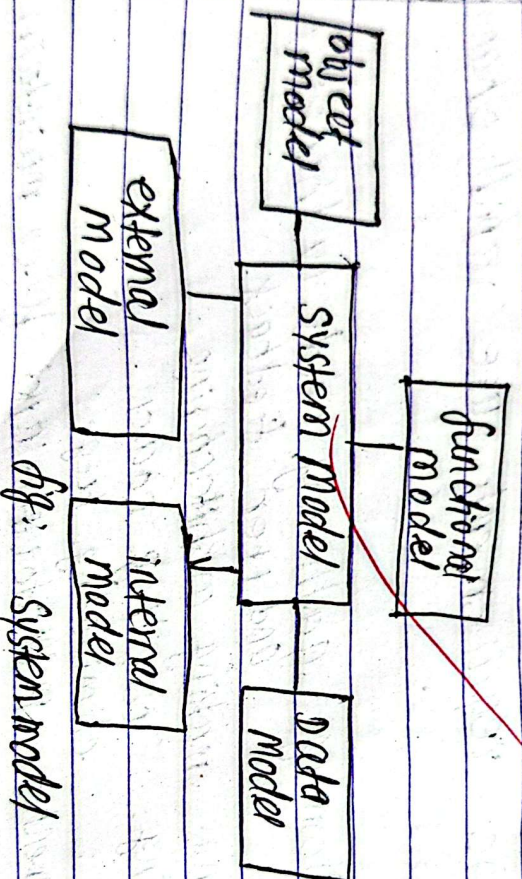


fig: system model

a) Object Model

The system which follows object oriented approach. Every entity is made as object.

b) Data Model

The system which is emphasized on data. Data is main component of this.

c) functional model

The system in which function is prioritized. The system which performs different functions.

d) External Model

The system model which is influenced by external sources like environment, experts suggestion etc.

e) Internal Model

Internal model is the model which is designed by internal needs of a individual, user or developer or organization to fulfill their requirements. It is based on the resources that are available in a organization.

7a Software Risk Management

Software risks are the problems arise in software development which decreases the quality of a software.

Software Risk Management is the process of defining, analysing, planning, and mitigating risks, which results a high quality software.

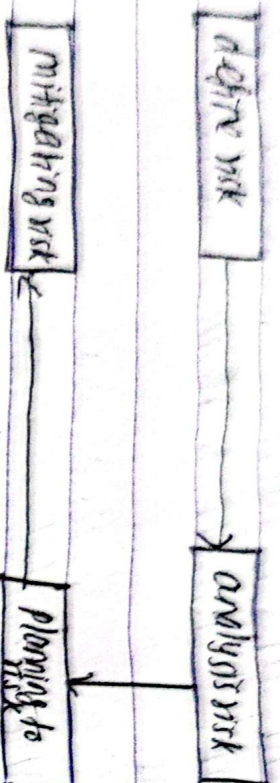


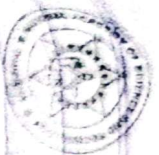
fig: Process of Risk Management of Software

i) Define risk

It defines the problem created in software and gathers them.

ii) Analysis risk

It analyses the risk how, why they come and how it affects software.



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iii) Planning to risk

It is the effective way to solve the risk.
It collects the required resource and budget to solve the risks.

iv) Mitigating risk

Based on planning risk are reduced using proper resource and provide problemless product.

Advantages

- high quality software
- risk handled
- user satisfactions

7b.

Incremental Software Development Model

- It is a famous SDLC model for large project.

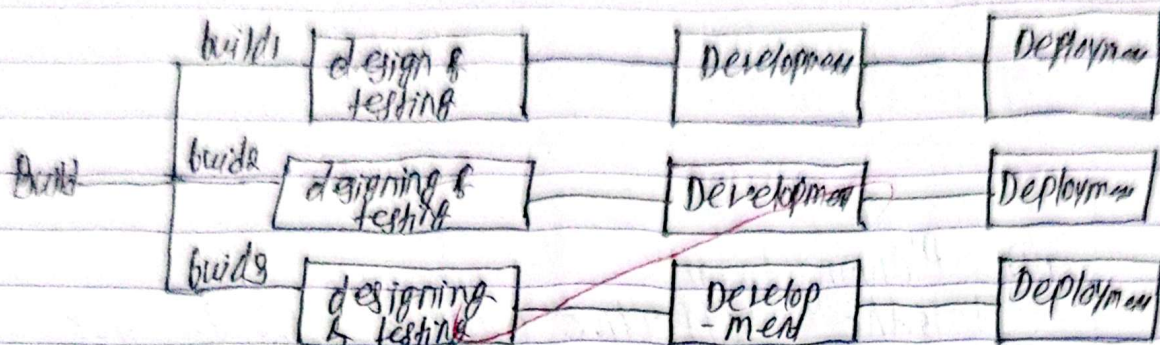


fig: Incremental software Development Model

- Incremental subdivides a development process.
- Here build1, build2 and build3 different develop software.
- Designing & testing, development and deploying are parallel in 3 different builds.

Advantages:

- parallel working
- fast and efficient

Disadvantages

- time consuming
 - requires more resources.
-